

Epochs '26 – Assignment 6 is Live!

It's time to put your **classification** skills into practice! 

In this challenge, you'll build and compare multiple Machine Learning models to predict whether a customer is likely to **churn**, helping businesses make smarter customer retention decisions.

 **Dataset:** Customer Churn Dataset

 <https://www.kaggle.com/datasets/muhammadshahidazeem/customer-churn-dataset>

or



<https://drive.google.com/drive/folders/15tsZ4P1itlWND6Z5P9PA44PXA1tYcEZx?usp=sharing>

What You'll Do

- ◆ Understand the business problem and identify the target variable.
- ◆ Prepare the data by:
 - Handling missing values
 - Encoding categorical features
 - Applying feature scaling (where required)
 - Splitting the dataset for training and testing
- ◆ Build and compare **at least three (3)** classification models.

Recommended Models:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- ◆ Evaluate each model using:
 - Accuracy
 - Precision
 - Recall
 - F1-Score
 - Confusion Matrix
- ◆ Compare the models, identify the best-performing one, and justify your choice.

Submission

Your GitHub repository should contain:

 `customer_churn_prediction.ipynb`

- Data preprocessing
- Model development
- Model evaluation
- Model comparison
- Final model selection

 `README.md`

- Participant Name
- MUID
- Business objective
- Dataset overview
- Features & target variable
- Preprocessing pipeline
- Models implemented
- Performance comparison
- Best model with justification
- Key observations
- Business recommendations
- Future improvements

Note: Please include your **Name** and **MUID** at the top of your README .md.

Submit your repository link in **#epochs-submissions** using the tag:

`#evn-ds-epochs26-day06`

★ **Reward:** 200 Karma Points

🕒 **Deadline:** Within **48 hours** after the completion of the **Day 7** session.